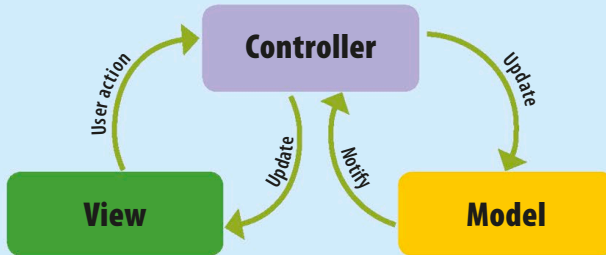


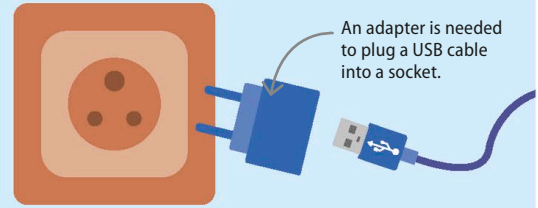
Examples of design patterns

Design patterns have become an industry standard. Most big programs have this ability to “snap off” certain behaviours and “snap in” the replacement. However, patterns aren’t a perfect solution, as each one has its drawbacks. Some programmers are critical of design patterns because they can sometimes be unnecessary, and not be the right fit for the problem at hand.



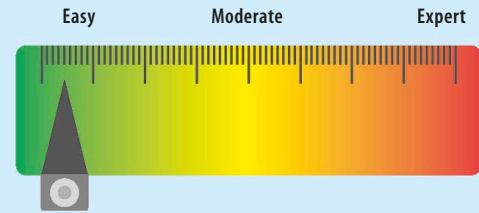
△ Model-view-controller

The model-view-controller design pattern is incredibly common – for example, it is used by every web browser. Essentially, it splits an application into three interconnected parts. The model is the base information. It is akin to HTML code for a webpage. The view is the visual representation of the model. For webpages, this is akin to CSS. The controller is the link between the user and the system – the way the information is shown, which is what the web browser does.



△ Adapter (or Wrapper, Translator)

Adapters bridge the gap when the format of data produced by one source does not work with the format required by another program or part of a program. The adapter takes the input data and outputs it in a way that can be used. This is similar to the way an adapter plug converts one type of plug to another.



△ Strategy

Computer games often have different difficulty levels. The Strategy design pattern makes it easier for different code to be written and organized so the player can pick the AI's difficulty level.

Planning for things to go wrong

Developers imagine how their software could be misused: people with long usernames, clients who sign up with no password, people who click the wrong button. To handle this scenario they might think of displaying an error message, or may bring the user back to the main page, or even make the program crash.

▷ Input validation

This means checking the data provided by the user. If a telephone number is required, it should only contain digits, no letters or symbols. This prevents crashes down the line.

First name	Last name	
James	Carter	
Street address		Apt, suite, bldg.
Ned Avenue		28
Country	City	State
United States	Malibu	California
Phone number		
434-343-328A		
This entry can only contain numbers.		
☐		

IN DEPTH

Integrated user feedback

As technology evolves, so does the development process. With the popularity of smartphones and the increase in internet speeds, more software is becoming downloadable. It's also easier for companies to get feedback. Products tend to be released quickly and updated continuously, as developers integrate user comments into new features. This means that software is becoming more responsive and more collaborative. Customers get the exact features they want.